

SRI CHAITANYA IIT ACADEMY, INDIA

AP • TELANGANA • KARNATAKA • TAMILNADU • MAHARASHTRA • DELHI • RANCHI

PRAGYA G8 – ACHIEVERS | MATHEMATICS
INTEGRATED PAPER II (HELL LEVEL) | JEE ADVANCED PATTERN*Every question draws on at least three of: Surds, Logarithms, Sequences & Series, Quadratic Equations***SECTION – I**
(INTEGER ANSWER TYPE)

- This section contains EIGHT (08) questions.
- The answer to each question is a SINGLE DIGIT INTEGER ranging from 0 TO 9, BOTH INCLUSIVE.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +3 if ONLY the correct option is chosen;
Zero Marks: 0 if the question is unanswered;
Negative Marks: –1 in all other cases.

1. Let $u = \sqrt[3]{7 + 5\sqrt{2}}$ and $v = \sqrt[3]{7 - 5\sqrt{2}}$. If $u + v$ and uv are the roots of $x^2 + px + q = 0$, and p, q, r are in geometric progression, then $|r|$ is

2. Let a, b, c be in arithmetic progression, b, c, d in geometric progression and c, d, e in harmonic progression, where $a = 1, e = 25$ and all of a, b, c, d, e are positive. Then c is

3. Let $x = \sqrt{12 + \sqrt{12 + \sqrt{12 + \cdots \infty}}}$ and $y = \sqrt{72 + \sqrt{72 + \sqrt{72 + \cdots \infty}}}$. If x, g, y are in geometric progression with $g > 0$, then $\log_g(xy)$ is

4. Given $\log_{10} 2 = 0.3010$ and $\log_{10} 3 = 0.4771$, let d be the number of digits in $(\sqrt{6})^{20}$ and let z be the number of zeros after the decimal point and before the first significant digit of $(\sqrt{6})^{-20}$. Then $d - z$ is

5. Let $\alpha > \beta$ be the roots of $x^2 - x\sqrt{9 + 4\sqrt{5}} + \sqrt{5} = 0$. If $\alpha - \beta$ is the second term of a geometric progression whose first term is $\frac{1}{3}$, then its positive common ratio is

6. Let $x_1 < x_2$ be the real values satisfying $(\log_5 x)^2 - \log_5(x^3) + 2 = 0$. If x_1, x_2, y are in geometric progression, then $\log_5 y + 4$ is

7. Let $a_1, a_2, \dots, a_7$ be in geometric progression with $a_1 = \sqrt{2}, a_7 = 8\sqrt{2}$ and positive common ratio. If a_4 is a root of $x^2 - kx + 8 = 0$ whose other root is also a term of this progression, then k is

8. Let T be the sum of all real values of x satisfying $(x^2 - 7x + 11)^{x^2 - 13x + 42} = 1$. If T is the third term of a geometric progression whose first term is 3 and whose common ratio r is positive, then $\log_r T$ is
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SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

- This section contains SIX (06) questions.
 - Each question has FOUR options (A), (B), (C) and (D). ONE OR MORE THAN ONE of these four options is(are) correct.
 - Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +4 ONLY if all the correct option(s) is(are) chosen;
Partial Marks: +3 if all the four options are correct but ONLY three options are chosen;
Partial Marks: +2 if three or more options are correct but ONLY two options are chosen, both of which are correct;
Partial Marks: +1 if two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks: 0 if none of the options is chosen;
Negative Marks: –2 in all other cases.
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9. Let $a = \sqrt{8 + 2\sqrt{15}}$ and $b = \sqrt{8 - 2\sqrt{15}}$. Then
- (A) $a + b = 2\sqrt{5}$
(B) $ab = 2$
(C) a and b are the roots of $x^2 - 2\sqrt{5}x + 2 = 0$
(D) $a, 2, b$ are in arithmetic progression
10. An infinite geometric progression has sum 5 and the sum of the squares of its terms is 15. Then
- (A) its common ratio is $\frac{1}{4}$
(B) its first term is $\frac{15}{4}$
(C) the sum of the cubes of its terms is $\frac{375}{13}$
(D) its third term is $\frac{15}{64}$
11. Let $x_1 > x_2$ be the real numbers satisfying $(\log_3 x)^2 - 4\log_3 x + 3 = 0$. Then
- (A) $\log_3 x_1 + \log_3 x_2 = 3$
(B) $x_1 x_2 = 81$
(C) $x_1, 9, x_2$ are in geometric progression
(D) $x_1 - x_2 = 24$
12. Let $p = \sqrt{14 + 6\sqrt{5}}$ and $q = \sqrt{14 - 6\sqrt{5}}$. Then
- (A) $p + q = 6$
(B) $pq = 2$
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- (C) $p, 2, q$ are in geometric progression
 (D) p and q are the roots of $x^2 - 6x + 4 = 0$
13. Let $\alpha > \beta$ be the roots of $x^2 - 6x + 7 = 0$. Then
 (A) $\alpha - \beta = 2\sqrt{2}$
 (B) $\alpha\beta = 7$
 (C) $\alpha + \beta, \alpha\beta, \alpha - \beta$ are in arithmetic progression
 (D) $\log_{\alpha} \beta = -1$
14. Let a, b, c be positive reals, none equal to 1, in geometric progression, and let $N > 0, N \neq 1$. Then
 (A) $\log_a N, \log_b N, \log_c N$ are in harmonic progression
 (B) a, b, c are in arithmetic progression
 (C) a^2, b^2, c^2 are in geometric progression
 (D) $\log_a N, \log_b N, \log_c N$ are in arithmetic progression

SECTION – III
(SINGLE CORRECT ANSWER TYPE)

- This section contains FOUR (04) questions.
- Each question has FOUR options (A), (B), (C) and (D). ONLY ONE of these four options is correct.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +3 if ONLY the correct option is chosen;
Zero Marks: 0 if the question is unanswered;
Negative Marks: -1 in all other cases.

15. Let $a, b > 0$ with $a \neq 1, b \neq 1$. If the equations $x^2 + (\log_a b)x + \log_b a = 0$ and $x^2 + (\log_b a)x + \log_a b = 0$ have both roots common and real, and $\alpha > \beta$ are those common roots, then $(\alpha - \beta) + ab$ equals
 (A) 2 (B) $\sqrt{5} - 1$
 (C) $2\sqrt{5}$ (D) $\sqrt{5} + 1$
16. Let a, b, c be in harmonic progression with $a = 7 + 4\sqrt{3}$ and $c = 7 - 4\sqrt{3}$. If b, k and $\log_{(2+\sqrt{3})} a$ are in arithmetic progression, then k equals
 (A) $\frac{15}{14}$ (B) $\frac{9}{7}$
 (C) $\frac{13}{14}$ (D) $\frac{8}{7}$
17. The harmonic mean of the roots of $x^2 - (\sqrt{5} + \sqrt{3})x + \sqrt{15} = 0$ is
 (A) $\sqrt{15}$ (B) $5\sqrt{3} - 3\sqrt{5}$
 (C) $\frac{\sqrt{5} + \sqrt{3}}{2}$ (D) $3\sqrt{5} - 5\sqrt{3}$

18. Let $x_1 > x_2$ be the real numbers satisfying $(\log_2 x)^2 = \log_2(x^2) + 8$. If x_1, g, x_2 are in geometric progression with $g > 0$, then g equals
- (A) 1 (B) 8
(C) 2 (D) 4